Linear Algebra - HW 2 Solutions

1) a)

	In	Out	Equation
A:	$300+500$	x_1+x_2	$x_1+x_2=800$
B:	x_2+x_4	x_3+300	$x_2-x_3+x_4=300$
C:	$400+100$	x_4+x_5	$x_4+x_5=500$
D:	x_1+x_5	600	$x_1+x_5=600$
All:	$300+500+100+400$	$300+x_3+600$	$1300=x_3+900$, $x_3=400$

System of equations:

$$\begin{cases} x_1+x_2 = 800 \\ x_2-x_3+x_4 = 300 \\ x_4+x_5 = 500 \\ x_1+x_5 = 600 \\ x_3 = 400 \end{cases}$$

matrix:

$$\begin{array}{c} \begin{array}{ccccc} x_1 & x_2 & x_3 & x_4 & x_5 \end{array} \\ \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 800 \\ 0 & 1 & -1 & 1 & 0 & 300 \\ 0 & 0 & 0 & 1 & 1 & 500 \\ 1 & 0 & 0 & 0 & 1 & 600 \\ 0 & 0 & 1 & 0 & 0 & 400 \end{array}\right] \end{array}$$

b) solve the above matrix to answer this!

you should get:

$x_1 = 600 - x_5$
$x_2 = 200 + x_5$
$x_3 = 400$
$x_4 = 500 - x_5$
$x_5 =$ free.

c) Nope! Since x_5 is free, we cannot find an exact answer. So, we can only give interval answers. Scenarios vary.

d) $0 \leq x_5 \leq 500$ ← min is 0 since we cannot have negative traffic volume, and the max is 500 since that's all the traffic coming into intersection C.

*we can use the bounds on x_5 to bound the other variables.

$0 \leq x_4 \leq 500$

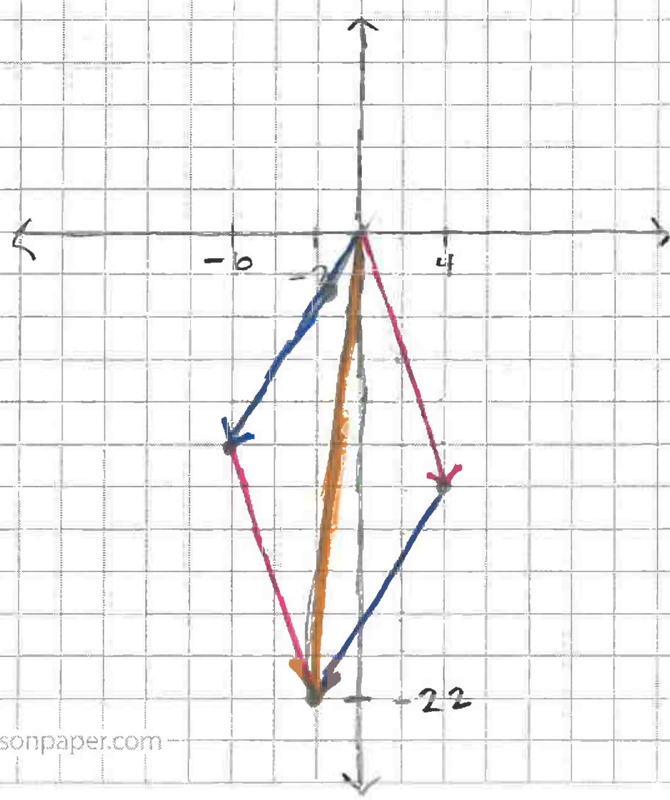
$400 \leq x_3 \leq 400$

$200 \leq x_2 \leq 700$

$100 \leq x_1 \leq 600$

2) $4\begin{bmatrix} 1 \\ -3 \end{bmatrix} - 2\begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \\ -12 \end{bmatrix} + \begin{bmatrix} -6 \\ -10 \end{bmatrix} = \begin{bmatrix} -2 \\ -22 \end{bmatrix}$

a)



b) The 4 represents the amount we "stretch" the vector $\begin{bmatrix} 1 \\ -3 \end{bmatrix}$. It will be 4 times as long as the original vector and in the same direction. The -2 represents a similar stretch (this time in the negative direction) but along the same line.

3) We would be able to reach Old Man Gauss's cabin if his location (107, 64) was a point on the line that passes through the origin and the vector point, but that is not the case!

→ our hoverboard only travels along the line $y = \frac{1}{3}x$ and $64 \neq \frac{107}{3}$

→ our magic carpet travels along the line $y = 2x$ and $64 \neq 2(107)$.

So, we need both modes of transport to reach Old Man Gauss.

4) We want to borrow the dragon (and not just because it's a dragon). Note that the jetpack's movement vector is a constant multiple of the magic carpet's movement vector, so they will move along the same line, thus restricting our movement to a linear space. The dragon's movement will allow us to reach Sophie's cottage.

$$\left[\begin{array}{cc|c} 1 & 5 & 76 \\ 2 & 1 & 95 \end{array}\right] \quad -2R_1 + R_2 \to R_2 \quad \left[\begin{array}{cc|c} 1 & 5 & 76 \\ 0 & -9 & -57 \end{array}\right]$$

$$-\frac{1}{9}R_2 \to R_2 \quad \left[\begin{array}{cc|c} 1 & 5 & 76 \\ 0 & 1 & 19/3 \end{array}\right]$$

$$-5R_2 + R_1 \to R_1 \quad \left[\begin{array}{cc|c} 1 & 0 & 133/3 \\ 0 & 1 & 19/3 \end{array}\right]$$

So, we can reach Sophie's cottage w/ our magic carpet (44.3 hrs) and the dragon (6.3 hours).

$$\frac{133}{3}\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \frac{19}{3}\begin{bmatrix} 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 76 \\ 95 \end{bmatrix}.$$